

Foundation (Jalapeno)

- Q1.** Which of the following is the vertex form of a quadratic function?
- (A) $y = a(x - h)^2 + k$
(B) $y = ax^2 + bx + c$
(C) $y = a(x + h)^2 - k$
(D) $y = ax^2 - bx - c$
(E) $y = a(x - h)(x - k)$
- Q2.** The vertex form $y = a(x - h)^2 + k$ has a vertex at (h, k) . What is the value of h if $y = 2(x - 3)^2 + 1$?
- (A) -3
(B) 3
(C) -2
(D) 2
(E) 1
- Q3.** Which of the following is the intercept form of a quadratic function?
- (A) $y = a(x - p)(x - q)$
(B) $y = ax^2 + bx + c$
(C) $y = a(x + p)(x + q)$
(D) $y = ax^2 - bx - c$
(E) $y = a(x - h)^2 + k$
- Q4.** The intercept form $y = a(x - p)(x - q)$ has x-intercepts at $(p, 0)$ and $(q, 0)$. What are the x-intercepts if $y = 2(x - 2)(x + 1)$?
- (A) $(-2, 0)$, $(1, 0)$
(B) $(2, 0)$, $(-1, 0)$
(C) $(-2, 0)$, $(-1, 0)$
(D) $(2, 0)$, $(1, 0)$
(E) $(-1, 0)$, $(2, 0)$
- Q5.** What is the value of a in the vertex form $y = a(x - h)^2 + k$ if $y = 3(x + 2)^2 - 1$?
- (A) -3
(B) 3
(C) -2
(D) 2
(E) 1
- Q6.** The quadratic function $y = x^2 - 4x - 3$ can be written in vertex form as $y = a(x - h)^2 + k$. What is the value of h ?
- (A) -2
(B) 2
(C) -1
(D) 1
(E) 3
- Q7.** The quadratic function $y = x^2 + 2x - 6$ can be written in intercept form as $y = a(x - p)(x - q)$. What are the values of p and q ?
- (A) $p = -3, q = 2$
(B) $p = 2, q = -3$
(C) $p = 1, q = -6$
(D) $p = -1, q = 6$
(E) $p = 3, q = -2$
- Q8.** What is the value of k in the vertex form $y = a(x - h)^2 + k$ if $y = 2(x + 1)^2 - 3$?
- (A) -3
(B) 3
(C) -2
(D) 2
(E) 1

Open Ended Questions (FRQ)

- Q9.** Write the quadratic function $y = x^2 + 5x + 6$ in vertex form $y = a(x - h)^2 + k$. Show all work and explain your reasoning.
- Q10.** Write the quadratic function $y = x^2 - 3x - 4$ in intercept form $y = a(x - p)(x - q)$. Identify the x-intercepts and explain their significance.

Chef's Tips

- Emphasize the importance of identifying the vertex and intercepts in quadratic functions.
- Use graphing calculators or software to visualize the quadratic functions and verify the vertex and intercepts.
- Provide feedback to students on the accuracy of their work and the clarity of their explanations.